CTP0007 Issue H Calculation Procedure

This document describes the calculation procedure implemented in `coupling_calculations.py` for the CTP0007 Issue H desktop application. Units are N, mm, MPa, and Nm unless otherwise noted. MPa is treated as N/mm2.

1. Input Selection

The screw type and size select a screw record from the embedded CTP 0007 lookup data. For `SPECIAL` screws, the user-entered geometry is used where applicable.

Material yield strength is selected from the material database unless the editable `TYS MPa` field is changed by the user.

Tightening input priority is:

User-entered tightening torque.

User-entered percent of tensile yield strength, `% TYS`.

Standard tightening torque from the selected screw type, size, and material.

If both tightening torque and `% TYS` are entered, the calculation is rejected.

2. Geometry

Definitions:

`d` = thread major diameter

`p` = thread pitch

`ds` = shank diameter

`dg` = groove diameter

`dc` = contact diameter

`Sy` = bolt tensile yield strength

`n` = screw count

`PCD` = pitch circle diameter

Thread pitch diameter:

dp = d - 0.6495 * p

Thread root diameter:

dr = d - 1.2268 * p

Tensile stress diameter:

dt = (dp + dr) / 2

Tensile stress area:

As = pi * ((dp + dr) / 4)^2

Shear and bending diameter:

dsb = dt when shear plane is Thread
dsb = ds when shear plane is Shank

Contact diameter:

dc = special contact diameter for SPECIAL screws
dc = special contact diameter when Nut contact is Special and value > 0
dc = standard record contact diameter otherwise

Average annular nut/contact radius:

Ro = dc / 2
Ri = dp / 2
Rn = (2 / 3) * ((Ro^3 - Ri^3) / (Ro^2 - Ri^2))

The contact diameter must be greater than the pitch diameter.

3. Effective Lever Arm

Effective shear/bending lever arm:

Leff = 0.05 mm for Stripper bolt
Leff = 0.15 * pack_thickness_mm for Drive bolt
Leff = 0.15 * pack_thickness_mm for Shim

Drive bolt and Shim require a positive pack thickness. Stripper bolt uses `N/A` pack thickness.

4. Tightening, Pretension, and Friction Torque

Friction definitions:

`mu_sn` = screw/nut friction coefficient

`mu_np` = nut/part friction coefficient

`mu_pp` = part/part friction coefficient

`Tt` = selected tightening torque

`Fv` = axial pretension

Axial pretension from tightening torque:

Fv = 1000 * Tt / (0.16 * p + 0.583 * mu_sn * dp + mu_np * Rn)

Axial pretension from `% TYS`:

Fv = (%TYS / 100) * As * Sy

Reported tightening torque when `% TYS` is used:

Tt_reported = (0.16 * p + 0.583 * mu_sn * dp + mu_np * Rn) * Fv / 1000

Preload percent of tensile yield strength:

preload_%TYS = Fv / (As * Sy) * 100

Nut/part friction torque component:

T_np = mu_np * Rn * Fv / 1000

Torsional tightening torque in the screw:

T_torsion = (0.16 * p + 0.583 * mu_sn * dp) * Fv / 1000

Flange friction torque capacity:

T_friction = Fv * n * mu_pp * PCD / 2 / 1000

Friction torque ratio is displayed as information:

ratio_% = T_friction / T_duty * 100

Friction torque lower than duty torque is not a warning by itself. Any uncovered torque is carried forward into the bolt shear/bending calculation.

5. Duty Torque Cases

The app evaluates three duty torque cases:

Continuous

Peak

Momentary

For each case:

T_residual = max(0, T_duty - T_friction)
V_total = 2000 * T_residual / (PCD * n)

`V_total` is the shear load per joint/bolt station.

If a sleeve is present and sleeve OD is greater than the thread major diameter:

V_sleeve = V_total * (OD^2 - d^2) / OD^2
V_bolt = V_total - V_sleeve

If no sleeve is present:

V_sleeve = 0
V_bolt = V_total

Sleeve type follows CTP0007 cell O33:

Sleeve type = No Sleeve when Sleeve OD is N/A/blank.

Sleeve type = Full Sleeve when Sleeve OD is present and shear plane is not Shank.

Sleeve type = Partial Sleeve when Sleeve OD is present and shear plane is Shank.

Full Sleeve and Partial Sleeve both affect sleeve shear share, sleeve stresses, and bolt residual shear share. Partial Sleeve additionally activates the shank total-shear check corresponding to CTP0007 rows P58:W60.

The app sleeve type input defaults to Auto, which follows the O33 spreadsheet rule above.

When the sleeve type is manually set to No Sleeve, Full Sleeve, or Partial Sleeve, the selected type overrides the O33-derived result. Manual Full Sleeve or Partial Sleeve requires a sleeve OD input.

6. Sleeve Stress and Safety Factor

Sleeve check is only numeric when `sleeve OD > d`. Otherwise the result is `No Sleeve`.

Sleeve bending stress:

$I\_sleeve = pi / 64 * (OD^4 - d^4)$
sigma_b_sleeve = V_sleeve * Leff * OD / 2 / I_sleeve

Sleeve shear stress, as implemented:

tau_sleeve = V_total / (pi / 4 * OD^2)

Sleeve Von Mises stress:

VM_sleeve = sqrt(sigma_b_sleeve^2 + 3 * tau_sleeve^2)

Sleeve yield safety factor:

SF_sleeve = sleeve_yield / VM_sleeve

Sleeve preload/gagging safety factor, matching CTP0007 cell AB34:

A_sleeve = max(0, pi / 4 * (OD^2 - d^2))

sigma_sleeve_preload = Fv / A_sleeve

SF_sleeve_preload = sleeve_yield / sigma_sleeve_preload

If no sleeve is fitted, the app reports `No Sleeve`. If sleeve OD is not greater than thread major diameter, the app reports `Gagging` to match the spreadsheet error branch.

Sleeve preload SF >= 1.25 minimum and >= 1.50 recommended.

Sleeve material options:

GMC 0401 = 245 MPa
GMC 0336 = 650 MPa
GMC0433 = 800 MPa

When sleeve OD is `N/A`, sleeve material and sleeve yield are `N/A`.

7. Bolt Shear/Bending Area Stress and Safety Factor

Bolt axial stress:

sigma_axial = 4 * Fv / (pi * dsb^2)

Bolt bending stress:

I_bolt = pi / 64 * dsb^4
sigma_b_bolt = V_bolt * Leff * dsb / 2 / I_bolt

Bolt shear stress:

tau_bolt = V_bolt / (pi / 4 * dsb^2)

Bolt maximum Von Mises stress:

VM_bolt = sqrt((sigma_axial + sigma_b_bolt)^2 + 3 * tau_bolt^2)

Bolt yield safety factor:

SF_bolt = Sy / VM_bolt

The table column `Maxi Stress (VM) MPa` reports `VM_bolt`.

Partial Sleeve shank total-shear check, active only when sleeve type is Partial Sleeve:

tau_partial_shank = V_total / (pi / 4 * dsb^2)

VM_partial_shank = sqrt(sigma_axial^2 + 3 * tau_partial_shank^2)

SF_partial_shank = Sy / VM_partial_shank

8. Assembly Stress in Groove Area

If `dg <= 0`, the result is `No groove`.

Groove axial stress:

sigma_groove = 4 * Fv / (pi * dg^2)

Groove twisting stress:

tau_groove = 16 * T_torsion * 1000 / (pi * dg^3)

Groove Von Mises stress:

VM_groove = sqrt(sigma_groove^2 + 3 * tau_groove^2)

Groove yield safety factor:

SF_groove = Sy / VM_groove

9. Assembly Stress in Thread Root Area

Thread-root axial stress:

sigma_thread = Fv / As

Thread-root twisting stress:

tau_thread = T_torsion * 1000 * dr / 2 / (pi / 32 * dr^4)

Thread-root Von Mises stress:

VM_thread = sqrt(sigma_thread^2 + 3 * tau_thread^2)

Thread-root yield safety factor:

SF_thread_root = Sy / VM_thread

10. Thread Pull-Out Stress

Thread engagement:

L = user-entered engagement
L = 1.2 * d when thread engagement mode is Thread/default

Thread pull-out area:

A_pullout = pi / 2 * dp * L

Engaged thread quantity:

N_engaged = L / p

First-thread load factor for `Emuge` or `Prevailing` screw/nut friction presets:

K = max(1,
0.001762 * N_engaged^3

- 0.028314 * N_engaged^2
+ 0.182016 * N_engaged
+ 0.720405)

First-thread load factor for other screw/nut friction presets:

K = max(1,
0.002317 * N_engaged^3
- 0.039254 * N_engaged^2
+ 0.416953 * N_engaged
+ 0.436158)

Thread pull-out stress:

sigma_pullout = Fv / A_pullout * K

The tapped-hole yield strength field is optional. If entered, tapped-hole yield must be greater than or equal to thread pull-out stress.

11. Minimum Safety Factor

The reported minimum safety factor is the minimum of all numeric safety factors:

minimum_SF = min(
all bolt shear/bending SF values,
all partial sleeve shank SF values when sleeve type is Partial Sleeve,
sleeve preload SF when numeric,
groove SF when groove exists,
thread-root SF,
all sleeve SF values when sleeve exists
)

12. Checking Criteria

12.1 API 671 5th Edition

Yield safety factor limits for bolt shear/bending and sleeve area:

Continuous torque: SF >= 1.50
Peak torque: SF >= 1.15
Momentary torque: SF >= 1.00

12.2 API 671 4th Edition

Yield safety factor limits for bolt shear/bending and sleeve area:

Continuous torque: SF >= 1.25
Peak torque: SF >= 1.15
Momentary torque: SF >= 1.00

12.3 Groove Area

SF_groove >= 1.10

12.4 Thread Root Area

SF_thread_root >= 1.00 required
SF_thread_root >= 1.10 preferred

The preferred value is used because residual torque can reduce by about 20 percent upon loading.

12.5 Tightening Preload Percent

Stripper bolt: preload_%TYS <= 60
Shim: preload_%TYS <= 60
Drive bolt: preload_%TYS <= 75

12.6 Thread Pull-Out

tapped_hole_yield >= sigma_pullout

If tapped-hole yield is blank or zero, the app reports the pull-out stress with a note rather than a pass/fail result.

13. Input Validation

The calculation rejects invalid input before solving:

`PCD > 0`

`screw_count >= 1`

`thread > 0`

`pitch > 0`

checking standard must be recognized

joint type must be recognized

pack thickness must be zero or positive

Drive bolt and Shim require positive pack thickness

contact diameter must be greater than zero

tightening torque and `% TYS` cannot both be entered

screw/nut, nut/part, and part/part friction coefficients must be greater than zero

sleeve OD must be zero or positive

tapped-hole yield must be zero or positive

lever arm must be zero or positive

14. Legacy Quick Friction Model Appendix

The module retains an older quick friction fastener model for compatibility and tests. It is not the main CTP0007 Issue H GUI workflow.

Effective friction radius:

r_eff = (inner_radius + outer_radius) / 2

when the radius model is uniform wear, otherwise:

r_eff = (2 / 3) * ((outer_radius^3 - inner_radius^3) /
(outer_radius^2 - inner_radius^2))

Design torque:

T_steady = transmitted_torque * service_factor
T_design = max(T_steady, cyclic_torque, transient_torque)

Residual pretension after preload loss:

F_residual = F_initial * (1 - preload_loss_percent / 100)

Slip torque capacity:

T_slip = mu * (F_residual * bolt_count) * r_eff * friction_interfaces / 1000
SF_slip = T_slip / T_design

Required clamp load:

F_required_total = T_design * 1000 / (mu * r_eff * friction_interfaces)
F_required_per_bolt = F_required_total / bolt_count
F_required_initial = F_required_per_bolt / (1 - preload_loss_percent / 100)

Bolt yield and proof loads:

F_yield = yield_strength * tensile_area
F_proof = proof_strength * tensile_area

Bolt/joint stiffness split:

phi = k_bolt / (k_bolt + k_joint)
F_bolt_service = F_residual + phi * F_axial
F_residual_service = F_residual - (1 - phi) * F_axial

Yield utilization values:

assembly_yield_utilization = F_initial / F_yield
service_yield_utilization = F_bolt_service / F_yield
required_preload_yield_utilization = F_required_initial / F_yield

Legacy quick model warnings include low service factor, transient/cyclic torque governing, slip capacity below demand, preload/yield utilization above the selected limit, preload above proof load, and joint separation.

15. Formula-by-Formula Spreadsheet Cell Cross-Reference

This table maps each individual formula or calculation rule in the procedure to the corresponding source cell(s) in CTP 0007.xltx. Formula references use the workbook provided at: C:\Users\jtpkyyu\Smiths Group\Engineering Couplings - Coupling Technical Programs (CTP)\CTP 0007.xltx

Where the Issue H app adds behavior that is not present as a direct spreadsheet cell, the reference is marked as an app extension and the nearest source spreadsheet result cell is identified where applicable.

Procedure item	Formula or rule in procedure	CTP 0007.xltx cell reference	Cross-reference note
Source workbook	Reference workbook used for cell mapping	C:\Users\jtpkyyu\Smiths Group\Engineering Couplings - Coupling Technical Programs (CTP)\CTP 0007.xltx	Sheets referenced below are in this workbook unless noted otherwise.
Notation: d	d = thread major diameter	CTP 0007!U22; source selector A14/E14; manual SPECIAL input A16; lookup Donn?es Vis!B3:I168	U22 formula returns SPECIAL manual thread or database thread by Type and Size.
Notation: p	p = thread pitch	CTP 0007!U23; manual SPECIAL input E16; lookup Donn?es Vis!B3:I168	U23 formula returns SPECIAL manual pitch or database pitch.
Notation: ds	ds = shank diameter	CTP 0007!W22; manual SPECIAL input X16; lookup Donn?es Vis!B3:I168	Used when Shear plane is Shank.
Notation: dg	dg = groove diameter	CTP 0007!U30; manual SPECIAL input N16; lookup Donn?es Vis!B3:I168	Zero/blank/N/A means no groove in the app.
Notation: dc	dc = contact diameter	CTP 0007!U31; Standard/Special selector J31; special input P31; manual SPECIAL input R16; lookup Donn?es Vis!B3:I168	Equivalent to app Standard contact dia, Nut contact, and Special contact dia controls.
Notation: Sy	Sy = bolt tensile yield strength	CTP 0007!U35; material selector R14; manual override X14; material table Donn?es Vis!Z3:AA15	The app field TYS MPa maps to manual override X14 behavior.
Notation: n	n = screw count	CTP 0007!N14	Direct user input.
Notation: PCD	PCD = pitch circle diameter	CTP 0007!I14	Direct user input.
Thread pitch diameter	dp = d - 0.6495 * p	CTP 0007!U24	Spreadsheet formula: U22 - 0.6495 * U23.
Thread root diameter	dr = d - 1.2268 * p	CTP 0007!U25	Spreadsheet formula: U22 - 1.2268 * U23.
Tensile stress diameter	dt = (dp + dr) / 2	CTP 0007!U26	Spreadsheet formula: (U24 + U25) / 2.
Tensile stress area	As = pi * ((dp + dr) / 4)^2	CTP 0007!U27	Spreadsheet formula: ((U24 + U25) / 4)^2 * PI().
Pitch diameter area	A_dp = pi / 4 * dp^2	CTP 0007!AB24	Spreadsheet support area for pitch diameter.
Root diameter area	A_dr = pi / 4 * dr^2	CTP 0007!AB25	Spreadsheet support area for root diameter.
Shear/bending diameter, thread plane	dsb = dt when shear plane is Thread	CTP 0007!O28 and U28	U28 selects average of U24 and U25 when O28 = Thread.
Shear/bending diameter, shank plane	dsb = ds when shear plane is Shank	CTP 0007!O28 and U28; shank source W22 / Donn?es Vis!B3:I168	U28 selects shank lookup when O28 = Shank.
Shear/bending area	A_dsb = pi / 4 * dsb^2	CTP 0007!AB28	Spreadsheet formula: PI()/4 * U28^2.
Contact diameter, SPECIAL screw	dc = special contact diameter for SPECIAL screws	CTP 0007!U31 using A14, R16	App uses entered Standard contact dia for SPECIAL geometry.
Contact diameter, special nut contact	dc = special contact diameter when Nut contact is Special and value > 0	CTP 0007!J31, P31, U31	J31/P31 select special contact diameter in the spreadsheet.
Contact diameter, standard	dc = standard record contact diameter otherwise	CTP 0007!U31; lookup Donn?es Vis!B3:I168	This is the default standard contact diameter.
Contact outer radius	Ro = dc / 2	Embedded in CTP 0007!U32	U32 uses U31/2 as the outside radius.
Contact inner radius	Ri = dp / 2	Embedded in CTP 0007!U32	U32 uses U24/2 as the inside radius.
Average annular nut radius	Rn = (2 / 3) * ((Ro^3 - Ri^3) / (Ro^2 - Ri^2))	CTP 0007!U32	Spreadsheet formula uses U31 and U24 directly.
Contact validation	dc must be greater than dp	No explicit original spreadsheet check cell; U32 becomes invalid if geometry is impossible	The app validates this before calculation.
Effective lever arm, spreadsheet	Leff = user-entered value in original spreadsheet	CTP 0007!U29	Original CTP0007 has direct Leff input.
Effective lever arm, stripper bolt	Leff = 0.05 mm for Stripper bolt	Derived app rule; spreadsheet equivalent value is entered in CTP 0007!U29	App extension requested for Issue H workflow.
Effective lever arm, drive bolt	Leff = 0.15 * pack_thickness_mm for Drive bolt	Derived app rule; spreadsheet equivalent value is entered in CTP 0007!U29	No original pack-thickness formula cell.
Effective lever arm, shim	Leff = 0.15 * pack_thickness_mm for Shim	Derived app rule; spreadsheet equivalent value is entered in CTP 0007!U29	No original pack-thickness formula cell.
Screw/nut friction coefficient	mu_sn from preset or custom value	CTP 0007!A19; preset selector A20; custom AD2; preset table Donn?es Vis!Z34:AA49	A19 formula applies custom when A20 selects Custom.
Nut/part friction coefficient	mu_np from preset or custom value	CTP 0007!D19; preset selector D20; custom AE2; preset table Donn?es Vis!Z34:AA49	D19 formula applies custom when D20 selects Custom.
Part/part friction coefficient	mu_pp from preset or custom value	CTP 0007!G19; preset selector G20; custom AF2; preset table Donn?es Vis!Z34:AA49	G19 formula applies custom when G20 selects Custom.
Continuous duty torque	T_duty_continuous = continuous torque input	CTP 0007!J19	Direct input in spreadsheet and app.
Peak duty torque	T_duty_peak = peak torque input	CTP 0007!O19	Template formula O19 = J19 * 2; app allows direct entry.
Momentary duty torque	T_duty_momentary = momentary torque input	CTP 0007!R19	Template formula R19 = O19 * 1.15; app allows direct entry.
Standard tightening torque	Tt = standard torque when standard torque is selected and no manual torque/%TYS is entered	CTP 0007!U37; Donn?es Vis!O3:X160; CTM 33 30 10!B15 and B18	U37 indexes the selected screw/material standard torque. CTM sheet is reference calculation.
Axial pretension from tightening torque	Fv = 1000 * Tt / (0.16 * p + 0.583 * mu_sn * dp + mu_np * Rn)	CTP 0007!U36 when V19 <> 0 or U37 is defined	Spreadsheet formula branch uses V19 or U37 as torque basis.
Axial pretension from %TYS	Fv = (%TYS / 100) * As * Sy	CTP 0007!U36 when V19 = 0 and Z19 <> 0	Spreadsheet formula branch uses Z19 * U27 * U35 / 100.
Reported tightening torque from %TYS	Tt_reported = (0.16 * p + 0.583 * mu_sn * dp + mu_np * Rn) * Fv / 1000	CTP 0007!U38 when Z19 <> 0	U38 reports the torque corresponding to the selected %TYS basis.
Preload percent of TYS	preload_%TYS = Fv / (As * Sy) * 100	CTP 0007!AD24 and message A38	AD24 is the ratio U36/U27/U35. A38 displays the same ratio as a percent.
Nut/part friction torque	T_np = mu_np * Rn * Fv / 1000	CTP 0007!U39	Spreadsheet formula: D19 * U32 * U36 / 1000.
Torsional tightening torque	T_torsion = (0.16 * p + 0.583 * mu_sn * dp) * Fv / 1000	CTP 0007!U40	Used by groove and thread-root assembly stress checks.
Flange friction torque	T_friction = Fv * n * mu_pp * PCD / 2 / 1000	CTP 0007!P43	Spreadsheet stores the common friction torque capacity in row 43.
Friction ratio, continuous	ratio_continuous = T_friction / T_duty_continuous	CTP 0007!P44	App displays this as percent.
Friction ratio, peak	ratio_peak = T_friction / T_duty_peak	CTP 0007!S44	App displays this as percent.
Friction ratio, momentary	ratio_momentary = T_friction / T_duty_momentary	CTP 0007!W44	App displays this as percent.
Residual torque, continuous	T_residual = max(0, T_duty - T_friction)	CTP 0007!P45	Spreadsheet uses IF(J19 < P43, 0, J19 - P43).

Procedure item	Formula or rule in procedure	CTP 0007.xltx cell reference	Cross-reference note
Residual torque, peak	T_residual = max(0, T_duty - T_friction)	CTP 0007!S45	Spreadsheet uses IF(O19 < P43, 0, O19 - P43).
Residual torque, momentary	T_residual = max(0, T_duty - T_friction)	CTP 0007!W45	Spreadsheet uses IF(R19 < P43, 0, R19 - P43).
Shear load per joint, continuous	V_total = 2000 * T_residual / (PCD * n)	CTP 0007!P46	Uses P45, I14, and N14.
Shear load per joint, peak	V_total = 2000 * T_residual / (PCD * n)	CTP 0007!S46	Uses S45, I14, and N14.
Shear load per joint, momentary	V_total = 2000 * T_residual / (PCD * n)	CTP 0007!W46	Uses W45, I14, and N14.
Sleeve shear share, continuous	V_sleeve = V_total * (OD^2 - d^2) / OD^2	CTP 0007!P49	Spreadsheet formula uses U33 sleeve OD and U22 thread major diameter.
Sleeve shear share, peak	V_sleeve = V_total * (OD^2 - d^2) / OD^2	CTP 0007!S49	Same formula for peak case.
Sleeve shear share, momentary	V_sleeve = V_total * (OD^2 - d^2) / OD^2	CTP 0007!W49	Same formula for momentary case.
Bolt shear share, continuous	V_bolt = V_total - V_sleeve	CTP 0007!P56	Spreadsheet formula: P46 - P49.
Bolt shear share, peak	V_bolt = V_total - V_sleeve	CTP 0007!S56	Spreadsheet formula: S46 - S49.
Bolt shear share, momentary	V_bolt = V_total - V_sleeve	CTP 0007!W56	Spreadsheet formula: W46 - W49.
No-sleeve branch	V_sleeve = 0 and V_bolt = V_total	CTP 0007!O33 plus P49:W49 and P56:W56	When sleeve OD is blank/zero/not active, P49:W49 become zero and bolt share remains total shear.
Sleeve type derivation	sleeve_type = No Sleeve, Full Sleeve, or Partial Sleeve	CTP 0007!O33	Spreadsheet formula: IF((U33>0)*(O28<>"Shank"), "Full", IF((U33=""), "No Sleeve", "Partial")). The app labels these as No Sleeve, Full Sleeve, and Partial Sleeve.
Sleeve type app override	Auto follows CTP 0007!O33; manual No Sleeve, Full Sleeve, or Partial Sleeve overrides the derived sleeve type	App extension; default source is CTP 0007!O33	Manual Full Sleeve or Partial Sleeve requires Sleeve OD. Manual No Sleeve disables sleeve share, sleeve preload, and partial shank checks even if an OD value remains entered.
No Sleeve type	Sleeve OD is blank/N/A; sleeve share and partial shank checks are inactive	CTP 0007!O33 plus P49:W49, P56:W56, P58:W60	App displays sleeve type No Sleeve and leaves sleeve/partial checks N/A where applicable.
Full Sleeve type	Sleeve OD is present and shear plane is not Shank	CTP 0007!O33; active rows P49:W53 and P56:W64	Sleeve share/stress and bolt residual shear/bending are active; partial shank rows P58:W60 are inactive.
Partial Sleeve type	Sleeve OD is present and shear plane is Shank	CTP 0007!O33; active rows P49:W53, P56:W64, and P58:W60	Partial Sleeve additionally activates the shank total-shear VM and SF rows P58:W60.
Sleeve second moment of area	I_sleeve = pi / 64 * (OD^4 - d^4)	Embedded in CTP 0007!P50/S50/W50; related detailed sheet Sleeve Bending!B17	Main sheet row 50 uses this denominator directly.
Sleeve bending stress, continuous	sigma_b_sleeve = V_sleeve * Leff * OD / 2 / I_sleeve	CTP 0007!P50	Uses P49, U29, U33, U22.
Sleeve bending stress, peak	sigma_b_sleeve = V_sleeve * Leff * OD / 2 / I_sleeve	CTP 0007!S50	Uses S49, U29, U33, U22.
Sleeve bending stress, momentary	sigma_b_sleeve = V_sleeve * Leff * OD / 2 / I_sleeve	CTP 0007!W50	Uses W49, U29, U33, U22.
Sleeve shear stress, continuous	tau_sleeve = V_total / (pi / 4 * OD^2)	CTP 0007!P51	Spreadsheet uses total joint shear P46 over sleeve OD area.
Sleeve shear stress, peak	tau_sleeve = V_total / (pi / 4 * OD^2)	CTP 0007!S51	Spreadsheet uses total joint shear S46 over sleeve OD area.
Sleeve shear stress, momentary	tau_sleeve = V_total / (pi / 4 * OD^2)	CTP 0007!W51	Spreadsheet uses total joint shear W46 over sleeve OD area.
Sleeve Von Mises, continuous	VM_sleeve = sqrt(sigma_b_sleeve^2 + 3 * tau_sleeve^2)	CTP 0007!P52	Main-sheet sleeve VM for continuous case.
Sleeve Von Mises, peak	VM_sleeve = sqrt(sigma_b_sleeve^2 + 3 * tau_sleeve^2)	CTP 0007!S52	Main-sheet sleeve VM for peak case.
Sleeve Von Mises, momentary	VM_sleeve = sqrt(sigma_b_sleeve^2 + 3 * tau_sleeve^2)	CTP 0007!W52	Main-sheet sleeve VM for momentary case.
Sleeve safety factor, continuous	SF_sleeve = sleeve_yield / VM_sleeve	CTP 0007!P53	Sleeve yield is CTP 0007!U34.
Sleeve safety factor, peak	SF_sleeve = sleeve_yield / VM_sleeve	CTP 0007!S53	Sleeve yield is CTP 0007!U34.
Sleeve safety factor, momentary	SF_sleeve = sleeve_yield / VM_sleeve	CTP 0007!W53	Sleeve yield is CTP 0007!U34.
Sleeve preload annulus area	A_sleeve = max(0, pi / 4 * (OD^2 - d^2))	CTP 0007!AB33	Spreadsheet formula: MAX(0, PI()/4*(U33^2-U22^2)).
Sleeve preload stress	sigma_sleeve_preload = Fv / A_sleeve	Embedded in CTP 0007!AB34 as U36/AB33	Uses axial pretension U36 and sleeve annulus area AB33.
Sleeve preload safety factor	SF_sleeve_preload = sleeve_yield / sigma_sleeve_preload	CTP 0007!AB34	Spreadsheet formula: IFERROR(U34/(U36/AB33), "Gagging"). The app displays No Sleeve when sleeve OD is N/A.
Sleeve preload SF criteria	SF_sleeve_preload >= 1.25 minimum; >= 1.50 recommended	App criterion using CTP 0007!AB34 equivalent result	Added to the app Stresses and Checks table and included in Minimum safety factor when numeric.
Detailed sleeve VM support	VM_sleeve_detail = max(S_VM along sampled beam)	Sleeve Bending!B28; sampled formulas Sleeve Bending!J49:J59	Used by the original detailed allowable peak calculation, not the app summary table.
Detailed sleeve SF support	SF_sleeve_detail = sleeve_yield / VM_sleeve_detail	Sleeve Bending!B30	Used by Sleeve Bending!B32 allowable peak.
Bolt axial stress	sigma_axial = 4 * Fv / (pi * dsb^2)	CTP 0007!P57/S57/W57	Same formula is repeated for all three duty columns.
Bolt second moment of area	I_bolt = pi / 64 * dsb^4	Embedded in CTP 0007!P61/S61/W61; related detailed sheet Bolt Bending!B17	Main sheet row 61 uses this denominator directly.
Bolt bending stress, continuous	sigma_b_bolt = V_bolt * Leff * dsb / 2 / I_bolt	CTP 0007!P61	Uses P56, U29, U28.
Bolt bending stress, peak	sigma_b_bolt = V_bolt * Leff * dsb / 2 / I_bolt	CTP 0007!S61	Uses S56, U29, U28.
Bolt bending stress, momentary	sigma_b_bolt = V_bolt * Leff * dsb / 2 / I_bolt	CTP 0007!W61	Uses W56, U29, U28.
Bolt shear stress, continuous	tau_bolt = V_bolt / (pi / 4 * dsb^2)	CTP 0007!P62	Uses P56 and U28.
Bolt shear stress, peak	tau_bolt = V_bolt / (pi / 4 * dsb^2)	CTP 0007!S62	Uses S56 and U28.

Procedure item	Formula or rule in procedure	CTP 0007.xltx cell reference	Cross-reference note
Bolt shear stress, momentary	tau_bolt = V_bolt / (pi / 4 * dsb^2)	CTP 0007!W62	Uses W56 and U28.
Bolt Von Mises, continuous	VM_bolt = sqrt((sigma_axial + sigma_b_bolt)^2 + 3 * tau_bolt^2)	CTP 0007!P63	Corresponds to app table Maxi Stress (VM) MPa for Continuous.
Bolt Von Mises, peak	VM_bolt = sqrt((sigma_axial + sigma_b_bolt)^2 + 3 * tau_bolt^2)	CTP 0007!S63	Corresponds to app table Maxi Stress (VM) MPa for Peak.
Bolt Von Mises, momentary	VM_bolt = sqrt((sigma_axial + sigma_b_bolt)^2 + 3 * tau_bolt^2)	CTP 0007!W63	Corresponds to app table Maxi Stress (VM) MPa for Momentary.
Bolt safety factor, continuous	SF_bolt = Sy / VM_bolt	CTP 0007!P64	Uses Sy from U35.
Bolt safety factor, peak	SF_bolt = Sy / VM_bolt	CTP 0007!S64	Uses Sy from U35.
Bolt safety factor, momentary	SF_bolt = Sy / VM_bolt	CTP 0007!W64	Uses Sy from U35.
Partial sleeve shank shear stress	tau_partial = V_total / (pi / 4 * dsb^2) when O33 = Partial	CTP 0007!P58/S58/W58	Spreadsheet partial-sleeve support block; app reports the main P64/S64/W64 equivalent bolt SF.
Partial sleeve shank VM	VM_partial = sqrt(sigma_axial^2 + 3 * tau_partial^2) when O33 = Partial	CTP 0007!P59/S59/W59	Spreadsheet partial-sleeve support block.
Partial sleeve shank SF	SF_partial = Sy / VM_partial	CTP 0007!P60/S60/W60	Spreadsheet partial-sleeve support block.
Detailed bolt VM support	VM_bolt_detail = max(S_VM along sampled beam)	Bolt Bending!B32; sampled formulas Bolt Bending!J49:J59	Used by original detailed allowable peak calculation.
Detailed bolt SF support	SF_bolt_detail = Sy / VM_bolt_detail	Bolt Bending!B34	Used by Bolt Bending!B36 allowable peak.
Groove axial stress	sigma_groove = 4 * Fv / (pi * dg^2)	CTP 0007!U67	Returns No groove when U30 = 0.
Groove twisting stress	tau_groove = 16 * T_torsion * 1000 / (pi * dg^3)	CTP 0007!U68	Uses U40 torsional tightening torque and U30 groove diameter.
Groove Von Mises	VM_groove = sqrt(sigma_groove^2 + 3 * tau_groove^2)	CTP 0007!U69	Blank when no groove.
Groove safety factor	SF_groove = Sy / VM_groove	CTP 0007!U70	Checked against 1.10 in app and spreadsheet check text.
Thread-root axial stress	sigma_thread = Fv / As	CTP 0007!U73	Uses U36 and U27.
Thread-root twisting stress	tau_thread = T_torsion * 1000 * dr / 2 / (pi / 32 * dr^4)	CTP 0007!U74	Uses U40 and U25.
Thread-root Von Mises	VM_thread = sqrt(sigma_thread^2 + 3 * tau_thread^2)	CTP 0007!U75	Main assembly stress in thread root area.
Thread-root safety factor	SF_thread_root = Sy / VM_thread	CTP 0007!U76	Spreadsheet label says mini 1.10; app also distinguishes required 1.00 and preferred 1.10.
Thread engagement, spreadsheet default	L = 1.2 * d	CTP 0007!U79	Spreadsheet formula U22 * 1.2.
Thread engagement, manual app mode	L = user-entered engagement	App extension; no original CTP 0007 input cell	Manual engagement overrides the spreadsheet default in the app.
Engaged thread quantity	N_engaged = L / p	CTP 0007!U80	Spreadsheet formula U79 / U23.
Thread pull-out area	A_pullout = pi / 2 * dp * L	CTP 0007!U81	Spreadsheet formula PI()/2 * U24 * U79.
First-thread load factor, Emuge/Prevailing	K = max(1, 0.001762*N^3 - 0.028314*N^2 + 0.182016*N + 0.720405)	Embedded in CTP 0007!U82	Spreadsheet chooses this branch when A20 contains Emuge or Prevailing.
First-thread load factor, other presets	K = max(1, 0.002317*N^3 - 0.039254*N^2 + 0.416953*N + 0.436158)	Embedded in CTP 0007!U82	Spreadsheet chooses this branch for other screw/nut friction presets.
Thread pull-out stress	sigma_pullout = Fv / A_pullout * K	CTP 0007!U82	Equivalent to U73 * U27 / U81 multiplied by the first-thread factor.
Tapped-hole yield comparison	tapped_hole_yield >= sigma_pullout	App extension; source stress is CTP 0007!U82	No original tapped-hole yield input cell in the spreadsheet.
Minimum safety factor	minimum_SF = min(active numeric SF values including sleeve SF, partial sleeve shank SF, sleeve preload SF, bolt SF, groove SF, and thread-root SF)	CTP 0007!AB19 plus app-added AB34 and P60/S60/W60 equivalent active checks	Spreadsheet AB19 uses P53:S53:W53, P64:S64:W64, U70, and U76. The app also includes sleeve preload AB34 and partial sleeve shank SF P60/S60/W60 when numeric.
Template continuous SF check	Continuous sleeve/bolt SF threshold in original template	CTP 0007!AB19 checks P53 < 1.5 and P64 < 1.5	Original template threshold.
Template peak SF check	Peak sleeve/bolt SF threshold in original template	CTP 0007!AB19 checks S53 < 1.25 and S64 < 1.25	Original template threshold.
Template momentary SF check	Momentary sleeve/bolt SF threshold in original template	CTP 0007!AB19 checks W53 < 1.1 and W64 < 1.1	Original template threshold.
App API671 5th continuous check	Continuous bolt/sleeve SF >= 1.50	App extension using CTP 0007!P53 and P64 equivalent results	No separate original standard selector cell.
App API671 5th peak check	Peak bolt/sleeve SF >= 1.15	App extension using CTP 0007!S53 and S64 equivalent results	No separate original standard selector cell.
App API671 5th momentary check	Momentary bolt/sleeve SF >= 1.00	App extension using CTP 0007!W53 and W64 equivalent results	No separate original standard selector cell.
App API671 4th continuous check	Continuous bolt/sleeve SF >= 1.25	App extension using CTP 0007!P53 and P64 equivalent results	No separate original standard selector cell.
App API671 4th peak check	Peak bolt/sleeve SF >= 1.15	App extension using CTP 0007!S53 and S64 equivalent results	No separate original standard selector cell.
App API671 4th momentary check	Momentary bolt/sleeve SF >= 1.00	App extension using CTP 0007!W53 and W64 equivalent results	No separate original standard selector cell.
Groove criterion	SF_groove >= 1.10	CTP 0007!A70 label and AB19 check on U70 < 1.1	Same threshold used by app.
Thread-root required criterion	SF_thread_root >= 1.00 required	App extension using CTP 0007!U76 equivalent result	Spreadsheet label/check uses 1.10 only.
Thread-root preferred criterion	SF_thread_root >= 1.10 preferred	CTP 0007!A76 label and AB19 check on U76 < 1.1	App warns below preferred but only fails required below 1.00.
Stripper bolt preload criterion	preload_%TYS <= 60	App extension; related preload ratio CTP 0007!AD24/A38	No original 60% cell.
Shim preload criterion	preload_%TYS <= 60	App extension; related preload ratio CTP 0007!AD24/A38	No original 60% cell.
Drive bolt preload criterion	preload_%TYS <= 75	CTM 33 30 10!B17 gives 75% reference utilization; related preload ratio CTP 0007!AD24/A38	App applies 75% maximum for drive bolt.
Input validation: PCD	PCD > 0	App validation; source input CTP 0007!I14	The spreadsheet expects a valid positive PCD input.
Input validation: screw count	screw_count >= 1	App validation; source input CTP 0007!N14	The spreadsheet expects a valid screw count.
Input validation: thread and pitch	thread > 0 and pitch > 0	App validation; source cells CTP 0007!U22 and U23	The spreadsheet formulas assume positive geometry.

Procedure item	Formula or rule in procedure	CTP 0007.xltx cell reference	Cross-reference note
Input validation: contact geometry	contact diameter > 0 and greater than pitch diameter	App validation; source cells CTP 0007!U31 and U24	Prevents invalid average radius calculation.
Input validation: tightening basis	do not enter tightening torque and %TYS together	App validation; source inputs CTP 0007!V19 and Z19	Spreadsheet branches between these inputs; app enforces one basis.
Input validation: friction coefficients	mu_sn, mu_np, mu_pp > 0	App validation; source cells CTP 0007!A19, D19, G19	Prevents invalid tightening/friction calculations.
Legacy quick model effective radius, wear	r_eff = (inner_radius + outer_radius) / 2	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model, not the CTP0007 Issue H spreadsheet procedure.
Legacy quick model effective radius, pressure	r_eff = (2/3) * ((outer_radius^3 - inner_radius^3) / (outer_radius^2 - inner_radius^2))	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model design torque	T_steady = transmitted_torque * service_factor; T_design = max(T_steady, cyclic_torque, transient_torque)	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model residual pretension	F_residual = F_initial * (1 - preload_loss_percent / 100)	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model slip capacity	T_slip = mu * (F_residual * bolt_count) * r_eff * friction_interfaces / 1000	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model slip SF	SF_slip = T_slip / T_design	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model required clamp total	F_required_total = T_design * 1000 / (mu * r_eff * friction_interfaces)	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model required clamp per bolt	F_required_per_bolt = F_required_total / bolt_count	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model required initial preload	F_required_initial = F_required_per_bolt / (1 - preload_loss_percent / 100)	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model yield load	F_yield = yield_strength * tensile_area	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model proof load	F_proof = proof_strength * tensile_area	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model load fraction	phi = k_bolt / (k_bolt + k_joint)	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model service bolt load	F_bolt_service = F_residual + phi * F_axial	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model service residual load	F_residual_service = F_residual - (1 - phi) * F_axial	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model assembly utilization	assembly_yield_utilization = F_initial / F_yield	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model service utilization	service_yield_utilization = F_bolt_service / F_yield	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.
Legacy quick model required preload utilization	required_preload_yield_utilization = F_required_initial / F_yield	No CTP 0007 spreadsheet cell	Legacy app-only compatibility model.